

Beyond Round Robin: A Comparative Evaluation of Machine Learning Models for Workload-Aware Mobile Financial Services Transaction Routing

Background

Mobile Financial Services systems process different types of transactions, such as balance inquiries, merchant payments, send-money requests, and high-value cash-outs. These transactions may require different levels of verification and computational resources.

Traditional routing methods such as Round Robin distribute transactions without understanding the processing requirements of individual transactions. This may cause unnecessary latency, inefficient infrastructure usage, and poor performance during traffic surges or node failures.

Problem Statement

Existing load-balancing methods mainly focus on server availability, CPU load, or request distribution. They may not consider the financial transaction's workload characteristics, risk indicators, required verification level, node capability, and processing cost together.

FinCluster AI aims to add a transaction-aware intelligent routing layer that classifies transactions as Heavy or Light workloads and routes them to suitable processing nodes.

Research Aim

The aim of this research is to compare multiple machine learning models for MFS transaction workload classification and examine how model choice affects routing latency, throughput, infrastructure cost, reliability, and fairness.

Proposed Models

- Logistic Regression
- Decision Tree
- Random Forest
- Extra Trees
- HistGradientBoosting
- XGBoost
- CatBoost

Optional later models:

- LightGBM
- Explainable Boosting Machine
- MLP

Routing Baselines

- Round Robin
- Weighted Round Robin
- Least-Loaded Routing
- Rule-Based Heavy/Light Routing
- ML Workload-Aware Routing
- ML + Node Health-Aware Routing

Proposed Scenarios

- Normal traffic
- Festival traffic surge
- Heavy/Light class imbalance
- Node warning
- Node failure
- Concept drift

Expected Contribution

This research will provide:

- A comparative benchmark of machine learning models for transaction workload classification.
- An evaluation of the operational impact of model selection on routing performance.
- A study of intelligent routing under traffic surges and node failures.
- A reproducible FinCluster simulation framework.

Current Limitations

- Synthetic or privacy-safe data
- Simulated infrastructure
- No real financial settlement
- No claim of production banking accuracy